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Metabolism

Ezetimibe

Ezetimibe is metabolized primarily in the small intestine and liver via glucuronide conjugation (a phase II reaction) with subsequent biliary excretion. Minimal oxidative metabolism (a phase I reaction) has been observed in all species evaluated. Ezetimibe and ezetimibe-glucuronide are the major drug-derived compounds detected in plasma, constituting approximately 10 to 20% and 80 to 90% of the total drug in plasma, respectively. Both ezetimibe and ezetimibe- glucuronide are slowly eliminated from plasma with evidence of significant enterohepatic recycling. The half-life for ezetimibe and ezetimibe-glucuronide is approximately 22 hours.

Simvastatin

Simvastatin is an inactive lactone which is readily hydrolyzed *in vivo* to the corresponding β -hydroxyacid, a potent inhibitor of HMG-CoA reductase. Hydrolysis takes place mainly in the liver; the rate of hydrolysis in human plasma is very slow.

In man simvastatin is well absorbed and undergoes extensive hepatic first-pass extraction. The extraction in the liver is dependent on the hepatic blood flow. The liver is its primary site of action, with subsequent excretion of drug equivalents in the bile. Consequently, availability of active drug to the systemic circulation is low.

Following an intravenous injection of the β -hydroxyacid metabolite, its half-life averaged 19 hours.

Elimination

Ezetimibe

Following oral administration of 14 C-ezetimibe (20 mg) to human subjects, total ezetimibe accounted for approximately 93% of the total radioactivity in plasma. Approximately 78% and 11% of the administered radioactivity were recovered in the feces and urine, respectively, over a 10-day collection period. After 48 hours, there were no detectable levels of radioactivity in the plasma.

Simvastatin

Following an oral dose of radioactive simvastatin to man, 13% of the radioactivity was excreted in the urine and 60% in the feces within 96 hours. The amount recovered in the feces represents absorbed drug equivalents excreted in bile as well as unabsorbed drug. Following an intravenous injection of the β -hydroxyacid metabolite an average of only 0.3% of the IV dose was excreted in urine as inhibitors.

Characteristics in Patients (Special Populations)

Paediatric Patients

The absorption and metabolism of ezetimibe are similar between children and adolescents (10 to 18 years) and adults. Based on total ezetimibe, there are no pharmacokinetic differences between adolescents and adults. Pharmacokinetic data in the pediatric population <10 years of age are not available.

Geriatric Patients

Plasma concentrations for total ezetimibe are about 2-fold higher in the elderly (≥ 65 years) than in the young (18 to 45 years). LDL-C reduction and safety profile are comparable between elderly and young subjects treated with ezetimibe.

Hepatic Insufficiency

After a single 10-mg dose of ezetimibe, the mean area under the curve (AUC) for total ezetimibe was increased approximately 1.7-fold in patients with mild hepatic insufficiency (Child-Pugh score 5 or 6), compared to healthy subjects. In a 14-day, multiple-dose study (10 mg daily) in patients with moderate hepatic insufficiency (Child-Pugh score 7 to 9), the mean AUC for total ezetimibe was increased approximately 4-fold on Day 1 and Day 14 compared to healthy subjects. No dosage adjustment is necessary for patients with mild hepatic insufficiency. Due to the unknown effects of the increased exposure to ezetimibe in patients with moderate or severe (Child-Pugh score ≥ 9) hepatic insufficiency, ezetimibe is not recommended in these patients.

Renal Insufficiency

Ezetimibe

After a single 10-mg dose of ezetimibe in patients with severe renal disease (n=8; mean CrCl ≤ 30 mL/min), the mean AUC for total ezetimibe was increased approximately 1.5-fold, compared to healthy subjects (n=9).

An additional patient in this study (post-renal transplant and receiving multiple medications, including cyclosporine) had a 12-fold greater exposure to total ezetimibe.

Simvastatin

In a study of patients with severe renal insufficiency (creatinine clearance < 30 mL/min), the plasma concentrations of total inhibitors after a single dose of a related HMG-CoA reductase inhibitor were approximately two-fold higher than those in healthy volunteers.

Gender

Plasma concentrations for total ezetimibe are slightly higher ($< 20\%$) in women than in men. LDL-C reduction and safety profile are comparable between men and women treated with ezetimibe.

Race

Based on a meta-analysis of pharmacokinetic studies with ezetimibe, there were no pharmacokinetic differences between Blacks and Caucasians.

Drug Interactions

Diltiazem

In a pharmacokinetic study, concomitant administration of diltiazem caused a 2.7-fold increase in exposure of simvastatin acid, presumably due to inhibition of CYP3A4.

Amlodipine

In a pharmacokinetic study, concomitant administration of amlodipine caused a 1.6-fold increase in exposure of simvastatin acid.

INDICATIONS

Prevention of Cardiovascular Events

XATHOR is indicated to reduce the risk of cardiovascular events (cardiovascular death, nonfatal myocardial infarction, nonfatal stroke, hospitalization for unstable angina, or

need for revascularization), in patients with coronary heart disease (CHD) and history of acute coronary syndrome (ACS), either previously treated with a statin or not.

Primary Hypercholesterolemia

XATHOR is indicated as adjunctive therapy to diet for the reduction of elevated total cholesterol (total-C), low-density lipoprotein cholesterol (LDL-C), apolipoprotein B (Apo B), triglycerides (TG), and non-high-density lipoprotein cholesterol (non-HDL-C), and to increase high-density lipoprotein cholesterol (HDL-C) in patients with primary (heterozygous familial and non-familial) hypercholesterolemia or mixed hyperlipidemia.

Homozygous Familial Hypercholesterolemia (HoFH)

XATHOR is indicated for the reduction of elevated total-C and LDL-C levels in patients with HoFH. Patients may also receive adjunctive treatments (e.g., LDL apheresis).

DOSEAGE AND ADMINISTRATION

The patient should be placed on a standard cholesterol-lowering diet before receiving XATHOR and should continue on this diet during treatment with XATHOR. The dosage should be individualized according to the baseline LDL-C level, the recommended goal of therapy, and the patient's response. XATHOR should be taken as a single daily dose in the evening, with or without food.

Patients with Primary Hypercholesterolemia

In patients with primary hyperlipidemia or mixed hyperlipidemia, the dosage range is 10/10 mg/day through 10/80 mg/day. The recommended usual starting dose is 10/20 mg/day. Initiation of therapy with 10/10 mg/day may be considered for patients requiring less aggressive LDL-C reductions. Patients who require a larger reduction in LDL-C (greater than 55%) may be started at 10/40 mg/day. After initiation or titration of XATHOR, lipid levels may be analyzed after 2 or more weeks and dosage adjusted, if needed. The 10/80 mg dose of ezetimibe/simvastatin is only recommended in patients at high risk for cardiovascular complications who have not achieved their treatment goals on lower doses and when the benefits are expected to outweigh the potential risks.

For patients who are currently on both simvastatin and ezetimibe, no dosage adjustment is required when switching to XATHOR.

Patients with Coronary Heart Disease

The starting dose may be initiated 10/40 mg once a day in the evening. The 10/80 mg dose of ezetimibe/simvastatin is only recommended when the benefits are expected to outweigh the potential risks. The 10/80mg dose is only recommended in patients at high risk for cardiovascular complications who have not achieved treatment goals on lower doses and when the benefits are expected to outweigh the potential risks.

Patients with Homozygous Familial Hypercholesterolemia

The recommended dosage for patients with homozygous familial hypercholesterolemia is XATHOR 10/40 mg/day or 10/80 mg/day in the evening. The 10/80 mg dose of ezetimibe/simvastatin is only recommended when the benefits are expected to outweigh the potential risks.

XATHOR should be used as an adjunct to other lipid-lowering treatments (e.g., LDL apheresis) in these patients or if such treatments are unavailable.

In patients taking lomitinapide concomitantly with XATHOR, the dose of XATHOR should not exceed 10/40 mg/day.

Use in the Elderly

No dosage adjustment is required for elderly patients.

Use in Paediatric Patients

Treatment with XATHOR is not recommended.

Use in Hepatic Impairment

No dosage adjustment is required in patients with mild hepatic insufficiency (Child-Pugh score 5 or 6). Treatment with XATHOR is not recommended in patients with moderate (Child-Pugh score 7 to 9) or severe (Child-Pugh score > 9) liver dysfunction.

Use in Renal Impairment

No dosage adjustment is required for patients with moderate renal insufficiency. If treatment in patients with severe renal insufficiency (creatinine clearance ≤ 30 mL/min) is deemed necessary, dosages above 10/10 mg/day should be implemented cautiously.

Coadministration with other medicines

Dosing of XATHOR should occur either ≥ 2 hours before or ≥ 4 hours after administration of a bile acid sequestrant.

In patients taking amiodarone, verapamil, diltiazem, or products containing elbasvir or grazoprevir concomitantly with XATHOR, the dose of Ezetimibe/Simvastatin should not exceed 10/20 mg/day.

In patients taking fibrates (other than gemfibrozil and fenofibrate) concomitantly with XATHOR, the dose of Ezetimibe/Simvastatin should not exceed 10/40 mg/day.

In patients taking amiodarone, verapamil or diltiazem concomitantly with XATHOR, the dose of Ezetimibe/Simvastatin should not exceed 10/20 mg/day.

In patients taking amlodipine or lipid-lowering dose of niacin (≥ 1 g/day) concomitantly with XATHOR, the dose of XATHOR should not exceed 10/40 mg/day.

The safety and effectiveness of XATHOR administered with fibrates have not been studied. Therefore, the combination of XATHOR and fibrates should be avoided.

XATHOR Tablet 10mg/40mg is not able to deliver all approved dose regimens of ezetimibe/ simvastatin. Therefore, other approved dosage forms and strengths of ezetimibe/simvastatin should be used in such cases.

ROUTE OF ADMINISTRATION

Oral.

CONTRAINDICATIONS

- Hypersensitivity to the active substances or to any of the excipients.
- Active liver disease or unexplained persistent elevations of serum transaminases.
- Pregnancy and nursing.
- Concomitant administration of potent CYP3A4 inhibitors (e.g., itraconazole, ketoconazole, posaconazole, voriconazole, HIV protease inhibitors, bocoprevir, telaprevir, erythromycin, clarithromycin, telithromycin, nefazodone, and drugs containing cobicistat).
- Concomitant administration of gemfibrozil, cyclosporine, or danazol.

PRECAUTIONS

Myopathy/Rhabdomyolysis

Simvastatin, like other inhibitors of HMG-CoA reductase, occasionally causes myopathy manifested as muscle pain, tenderness or weakness with creatine kinase (CK) above 10X the upper limit of normal (ULN). Myopathy sometimes takes the form of rhabdomyolysis with or without acute renal failure secondary to myoglobinuria, and rare fatalities have occurred. The risk of myopathy is increased by high levels of HMG-CoA reductase inhibitory activity in plasma (i.e., elevated simvastatin and simvastatin acid plasma levels), which may be due, in part, to interacting drugs that interfere with simvastatin metabolism and/or transporter pathways. Predisposing factors for myopathy include advanced age (≥ 65 years), female gender, uncontrolled hypothyroidism, and renal impairment.

As with other HMG-CoA reductase inhibitors, the risk of myopathy/rhabdomyolysis is dose related for simvastatin.

The risk of myopathy is greater in patients on simvastatin 80 mg compared with other statin- based therapies with similar LDL-C-lowering efficacy. Therefore, the 10/80 mg dose of Ezetimibe/Simvastatin should only be used in patients at high risk for cardiovascular complications who have not achieved their treatment goals on lower doses and when the benefits are expected to outweigh the potential risks. In patients taking Ezetimibe/Simvastatin 10/80 mg for whom an interacting agent is needed, a lower dose of XATHOR or an alternative statin-ezetimibe regimen with less potential for drug-drug interactions should be used.

All patients starting therapy with XATHOR, or whose dose of XATHOR is being increased, should be advised of the risk of myopathy and told to report promptly any unexplained muscle pain, tenderness or weakness. XATHOR therapy should be discontinued immediately if myopathy is diagnosed or suspected. The presence of these symptoms, and a CK level > 10 times the upper limit of normal indicates myopathy. In most cases, when patients were promptly discontinued from simvastatin treatment, muscle symptoms and CK increases resolved. Periodic CK determinations may be considered in patients starting therapy with XATHOR or whose dose is being increased. Periodic CK determinations are recommended for patients titrating to the 10/80 mg dose. There is no assurance that such monitoring will prevent myopathy.

Many of the patients who have developed rhabdomyolysis on therapy with simvastatin have had complicated medical histories, including renal insufficiency usually as a consequence of long-standing diabetes mellitus. Such patients taking XATHOR merit closer monitoring. Therapy with XATHOR should be temporarily stopped a few days prior to elective major surgery and when any major medical or surgical condition supervenes.

There have been very rare reports of an immune-mediated necrotizing myopathy (IMNM) during or after treatment with some statins. IMNM is clinically characterized by:

- persistent proximal muscle weakness and elevated serum creatine kinase, which persist despite discontinuation of statin treatment;
- muscle biopsy showing necrotizing myopathy without significant inflammation;
- improvement with immunosuppressive agents.

Drug Interactions

Because XATHOR contains simvastatin, the risk of myopathy/rhabdomyolysis is increased by concomitant use of XATHOR with the following drugs:

Contraindicated Drugs

- Potent inhibitors of CYP3A4: Concomitant use with medicines labeled as having a potent inhibitory effect on CYP3A4 at therapeutic doses (e.g., itraconazole, ketoconazole, posaconazole, voriconazole, erythromycin, clarithromycin, telithromycin, HIV protease inhibitors, bocoprevir, telaprevir, nefazodone, or drugs containing cobicistat) is contraindicated. If short-term treatment with potent CYP3A4 inhibitors is unavoidable, therapy with XATHOR should be suspended during the course of treatment.

- Gemfibrozil, cyclosporine, or danazol: Concomitant use of these drugs with XATHOR is contraindicated.

Other Drugs

- Fusicidic acid: Patients on fusicidic acid treated concomitantly with simvastatin may have an increased risk of myopathy/rhabdomyolysis. Coadministration with fusicidic acid is not recommended. In patients where the use of systemic fusicidic acid is considered essential, XATHOR should be discontinued throughout the duration of fusicidic acid treatment. In exceptional circumstances, where prolonged systemic fusicidic acid is needed, e.g. for the treatment of severe infections, the need for coadministration of XATHOR and fusicidic acid should only be considered on a case-by-case basis under close medical supervision.

- Amiodarone: The dose of Ezetimibe/Simvastatin should not exceed 10/20 mg daily in patients receiving concomitant medication with amiodarone.

- Calcium channel blockers

- Verapamil or diltiazem: Patients on diltiazem treated concomitantly with simvastatin 80 mg had an increased risk of myopathy. **The dose of Ezetimibe/Simvastatin should not exceed 10/20 mg daily in patients receiving concomitant medication with verapamil or diltiazem.**

- Amlodipine: Patients on amlodipine treated concomitantly with simvastatin 80 mg had a slightly increased risk of myopathy. **The dose of XATHOR should not exceed 10/40 mg daily in patients receiving concomitant medication with amlodipine.**

- Lomitapide: **The dose of XATHOR should not exceed 10/40 mg daily in patients with HoFH receiving concomitant medication with lomitapide.**

- Moderate inhibitors of CYP3A4: Patients taking other medicines labeled as having a moderate inhibitory effect on CYP3A4 concomitantly with XATHOR, particularly higher XATHOR doses, may have an increased risk of myopathy. When coadministering XATHOR with a moderate inhibitor of CYP3A4, a dose adjustment of XATHOR may be necessary.

- Inhibitors of Breast Cancer Resistant Protein (BCRP): Concomitant administration of products that are inhibitors of BCRP (e.g., elbasvir and grazoprevir) may lead to increased plasma concentrations of simvastatin and an increased risk of myopathy; therefore, a dose adjustment of XATHOR may be necessary. Coadministration of elbasvir and grazoprevir with simvastatin has not been studied; however, **the dose of Ezetimibe/Simvastatin should not exceed 10/20mg daily in patients receiving concomitant medication with products containing elbasvir or grazoprevir.**
- Other Fibrates: The safety and effectiveness of XATHOR administered with fibrates, have not been studied. **Therefore, the concomitant use of XATHOR and fibrates, should be avoided. Concomitant use of gemfibrozil is contraindicated.**
- Niacin (≥1 g/day): The benefit of the combined use of simvastatin with niacin should be carefully weighed against the potential risks of the combination. **While the only Asian population assessed in a clinical trial was Chinese, because the incidence of myopathy is higher in Chinese than in non-Chinese patients, coadministration of XATHOR with lipid- modifying doses (≥ 1 g/day) of niacin is not recommended in Asian patients.**
- Daptomycin: Reports of myopathy and/or rhabdomyolysis have been observed with HMG-CoA reductase inhibitors co-administered with daptomycin. Caution should be used when prescribing HMG-CoA reductase inhibitors with daptomycin, as either agent can cause myopathy and/or rhabdomyolysis when given alone. Consideration should be given to suspending XATHOR temporarily in patients taking daptomycin.
- Anticoagulants: If XATHOR is added to warfarin, another coumarin anticoagulant, or flutidione, the International Normalized Ratio (INR) should be appropriately monitored.

Myasthenia Gravis/Ocular Myasthenia

In few cases, statins have been reported to induce new onset or aggravate pre-existing myasthenia gravis or ocular myasthenia. XATHOR should be discontinued in case these conditions occur. There have been reports of recurrences of these conditions when the same or a different statin was (re-) administered.

Liver Enzymes

It is recommended that LFTs be performed before treatment with XATHOR begins and thereafter when clinically indicated. Patients titrated to the 10/80 mg dose should receive an additional test prior to titration, 3 months after titration to the 10/80 mg dose, and periodically thereafter (e.g., semiannually) for the first year of treatment. Special attention should be paid to patients who develop elevated serum transaminase levels, and in these patients, measurements should be repeated promptly and then performed more frequently. If the transaminase levels show evidence of progression, particularly if they rise to 3 X ULN and are persistent, the drug should be discontinued. Note that ALT may emanate from muscle, therefore ALT rising with CK may indicate myopathy.

There have been rare postmarketing reports of fatal and non-fatal hepatic failure in patients taking statins, including simvastatin. If serious liver injury with clinical symptoms and/or hyperbilirubinemia or jaundice occurs during treatment with XATHOR, promptly interrupt therapy. If an alternate etiology is not found do not restart XATHOR.

XATHOR should be used with caution in patients who consume substantial quantities of alcohol and/or have a past history of liver disease. Active liver diseases or unexplained persistent transaminase elevations are contraindications to the use of XATHOR.

Hepatic Insufficiency

Due to the unknown effects of the increased exposure to ezetimibe in patients with moderate or severe hepatic insufficiency, XATHOR is not recommended in these patients.

Excipients

Xathor tablets contain a sugar called lactose. If you have been told by your doctor that you have an intolerance to some sugars, contact your doctor before taking this medicinal product.

This medicine contains less than 1 mmol sodium (23mg) per tablet, that is to say essentially 'sodium-free'.

PREGNANCY

XATHOR is contraindicated during pregnancy.

Atherosclerosis is a chronic process, and ordinarily discontinuation of lipid-lowering drugs during pregnancy should have little impact on the long-term risk associated with primary hypercholesterolemia.

Simvastatin

The safety of simvastatin in pregnant women has not been established. No controlled clinical trials with simvastatin have been conducted in pregnant women. Rare reports of congenital anomalies following intrauterine exposure to HMG-CoA reductase inhibitors have been received.

XATHOR should not be used in women who are pregnant, trying to become pregnant or suspect they are pregnant. Treatment with XATHOR should be suspended for the duration of pregnancy or until it has been determined that the woman is not pregnant.

Ezetimibe

No clinical data on exposed pregnancies are available for ezetimibe.

When ezetimibe was given with simvastatin, no teratogenic effects were observed in embryo- fetal development studies in pregnant rats. In pregnant rabbits, a low incidence of skeletal malformations was observed.

NURSING MOTHERS

It is not known whether the active components of XATHOR are excreted into human breast milk; therefore, women who are nursing should not take XATHOR.

USE IN ELDERLY

Because advanced age (≥ 65 years) is a predisposing factor for myopathy, XATHOR should be prescribed with caution in the elderly. In a clinical trial of patients treated with simvastatin 80 mg/day, patients ≥ 65 years of age had an increased risk of myopathy compared to patients <65 years of age.

DRUG INTERACTIONS

XATHOR

No clinically significant pharmacokinetic interaction was seen when ezetimibe was coadministered with simvastatin.

XATHOR is bioequivalent to co-administered ezetimibe and simvastatin. Multiple mechanisms may contribute to potential interactions with HMG Co-A reductase inhibitors. Drugs or herbal products that inhibit certain enzymes (e.g. CYP3A4) and/or transporter (e.g. OATP1B) pathways may increase simvastatin and simvastatin acid plasma concentrations and may lead to an increased risk of myopathy/rhabdomyolysis.

Consult the prescribing information of all concomitantly used drugs to obtain further information about their potential interactions with simvastatin and/or the potential for enzyme or transporter alterations and possible adjustments to dose and regimens.

Contraindicated drugs

Concomitant use of the following drugs is contraindicated

Potent Inhibitors of CYP3A4

No clinically significant pharmacokinetic interactions have been observed between ezetimibe and drugs known to be metabolized by cytochromes P450 1A2, 2D6, 2C8, 2C9, and 3A4, or N-acetyltransferase. Simvastatin is metabolized by CYP3A4, but has no CYP3A4 inhibitory activity; therefore it is not expected to affect the plasma concentrations of other drugs metabolized by CYP3A4. Potent inhibitors of CYP3A4 increase the risk of myopathy by reducing the elimination of the simvastatin component of XATHOR: Concomitant use of drugs labeled as having a potent inhibitory effect on CYP3A4 at therapeutic doses (e.g., itraconazole, ketoconazole, posaconazole, voriconazole, erythromycin, clarithromycin, telithromycin, HIV protease inhibitors, boceprevir, telaprevir, nefazodone, drugs containing cobicistat) is contraindicated. If treatment with potent CYP3A4 inhibitors is unavoidable, therapy with simvastatin should be suspended during the course of treatment.

Gemfibrozil, Cyclosporine, or Danazol

Concomitant use of these drugs with simvastatin is contraindicated

Other drug interactions

Other Fibrates: The safety and effectiveness of XATHOR administered with fibrates have not been studied.

- Simvastatin - Fibrates can cause myopathy when given alone. Severe myositis with myoglobinuria has been reported with concomitant use of statins and fibrates. Therefore, the concomitant use of XATHOR and fibrates, should be avoided. Concomitant use of gemfibrozil is contraindicated. The dose of simvastatin should not exceed 10 mg daily in patients receiving concomitant medication with fibrates other than gemfibrozil or fenofibrate. When simvastatin and fenofibrate are given concomitantly, there is no evidence that the risk of myopathy exceeds the sum of the individual risks of each agent. Caution should be used when prescribing fenofibrate with simvastatin, as either agent can cause myopathy when given alone. Addition of fibrates to simvastatin typically provides little additional reduction in LDL-C, but further reductions of TG and further increases in HDLC may be obtained. Combinations of fibrates with simvastatin have been used without myopathy in small short-term clinical studies with careful monitoring.

- Ezetimibe – Fibrates may increase cholesterol excretion into the bile, leading to cholelithiasis. Coadministration of ezetimibe with other fibrates has not been studied.

Fusidic Acid: The risk of myopathy/rhabdomyolysis may be increased by concomitant administration of fusidic acid.

Amiodarone: The risk of myopathy/rhabdomyolysis is increased by concomitant administration of amiodarone with XATHOR. In a clinical trial, myopathy was reported in 6% of patients receiving simvastatin 80 mg and amiodarone. The dose of simvastatin should not exceed 20 mg daily in patients receiving concomitant medication with amiodarone.

Cholestyramine: Concomitant cholestyramine administration decreased the mean AUC of total ezetimibe (ezetimibe + ezetimibe glucuronide) approximately 55%. The incremental LDL-C reduction due to adding XATHOR to cholestyramine may be lessened by this interaction.

Calcium Channel Blockers: Verapamil or diltiazem: In a clinical trial, patients on diltiazem treated concomitantly with simvastatin 80mg had an increased risk of myopathy. The dose of simvastatin should not exceed 20mg daily in patients receiving concomitant medication with verapamil or diltiazem.

Amlodipine: In a clinical trial, patients on amlodipine treated concomitantly with simvastatin 80 mg had a slightly increased risk of myopathy. The dose of simvastatin should not exceed 40 mg daily in patients receiving concomitant medication with amlodipine.

Lomitapide: The risk of myopathy/rhabdomyolysis may be increased by concomitant administration of lomitapide.

Moderate Inhibitors of CYP3A4: Patients taking other medicines labeled as having a moderate inhibitory effect on CYP3A4 concomitantly with XATHOR, particularly higher XATHOR doses may have an increased risk of myopathy.

Inhibitors of the Transport Protein OATP1B: Simvastatin acid is a substrate of the transport protein OATP1B. Concomitant administration of medicinal products that are inhibitors of the transport protein OATP1B may lead to increased plasma concentrations of simvastatin acid and an increased risk of myopathy.

Inhibitors of Breast Cancer Resistant Protein (BCRP): Simvastatin is a substrate of the efflux transporter BCRP. Concomitant administration of products that are inhibitors of BCRP (e.g., elbasvir and grazoprevir) may lead to increased plasma concentrations of simvastatin and an increased risk of myopathy. When coadministering simvastatin with an inhibitor of BCRP, a dose adjustment of XATHOR may be necessary.

Niacin: The dose of simvastatin should not exceed 40mg daily in patients receiving concomitant medication with niacin (nicotinic acid) ≥ 1g/day. Cases of myopathy/rhabdomyolysis have been observed with simvastatin coadministered with lipid-modifying doses (≥ 1 g/day) of niacin.

Colchicine: There have been reports of myopathy and rhabdomyolysis with the concomitant administration of colchicine and XATHOR in patients with renal insufficiency. Close clinical monitoring of such patients taking this combination is advised.

Daptomycin: The risk of myopathy and/or rhabdomyolysis may be increased by concomitant administration of HMG-CoA reductase inhibitors and daptomycin.

Other Interactions

Grapefruit juice contains one or more components that inhibit CYP3A4 and can increase the plasma levels of drugs metabolized by CYP3A4. The effect of typical consumption (one 250mL glass daily) is minimal (13% increase in active plasma HMG-CoA reductase inhibitory activity as measured by the area under the concentration-time curve) and of no clinical relevance. However, because larger quantities significantly increase the plasma levels of HMG-CoA reductase inhibitory activity, grapefruit juice should be avoided during XATHOR therapy.

Anticoagulants

In patients taking coumarin anticoagulants, prothrombin time should be determined before starting XATHOR and frequently enough during early therapy to ensure that no significant alteration of prothrombin time occurs. Once a stable prothrombin time has been documented, prothrombin times can be monitored at the intervals usually recommended for patients on coumarin anticoagulants. If the dose of XATHOR is changed or discontinued, the same procedure should be repeated. Simvastatin therapy has not been associated with bleeding or with changes in prothrombin time in patients not taking anticoagulants.

The effect of XATHOR on the prothrombin time has not been studied.

Antacids: Concomitant antacid administration decreased the rate of absorption of ezetimibe but had no effect on the bioavailability of ezetimibe. This decreased rate of absorption is not considered clinically significant.

SIDE EFFECTS

XATHOR was generally well tolerated.

The following common (≥1/100, <1/10) or uncommon (≥1/1000, <1/100); drug-related adverse experiences were reported in patients taking ezetimibe/simvastatin and placebo:

Investigations:

Common: ALT and/or AST increased; blood CK increased
Uncommon: blood bilirubin increased; blood uric acid increased; gamma-glutamyltransferase increased; international normalised ratio increased; protein urine present; weight decreased

Nervous system disorders:

Uncommon: dizziness; headache

Gastrointestinal disorders:

Uncommon: abdominal pain; abdominal discomfort; abdominal pain upper; dyspepsia; flatulence; nausea; vomiting

Skin and subcutaneous tissue disorders:

Uncommon: pruritus; rash

Musculoskeletal and connective tissue disorders:

Uncommon: arthralgia; muscle spasms; muscular weakness; musculoskeletal discomfort; neckpain; pain in extremity
Not known : Immune-mediated necrotizing myopathy

General disorders and administration site conditions:

Uncommon: asthenia; fatigue; malaise; edema peripheral

Psychiatric disorders:

Uncommon: sleep disorders

Increases in HbA1c and fasting blood glucose have been reported with statins. The risk of hyperglycemia, however, is outweighed by the reduction in vascular risk with statins.

The following common (≥ 1/100, <1/10) or uncommon (≥1/1000, <1/100); drug-related adverse experiences were reported in patients taking ezetimibe/simvastatin and at a greater incidence than statins administered alone:

Investigations:

Common: ALT and/or AST increased
Uncommon: blood bilirubin increased; blood CK increased; gamma-glutamyltransferase increased

Nervous system disorders:

Uncommon: headache; paresthesia

Gastrointestinal disorders:

Uncommon: abdominal distension; diarrhea; dry mouth; dyspepsia; flatulence; gastroesophageal reflux disease; vomiting

Skin and subcutaneous tissue disorders:

Uncommon: pruritus; rash; urticaria

Musculoskeletal and connective tissue disorders:

Common: myalgia
Uncommon: arthralgia; back pain; muscle spasms; muscular weakness; musculoskeletal pain; pain in extremity

General disorders and administration site conditions:

Uncommon: asthenia; chest pain; fatigue; edema peripheral

Psychiatric disorders:

Uncommon: insomnia

Post-marketing Experience

The following additional adverse reactions have been reported in post-marketing use with ezetimibe/simvastatin or during clinical studies or post-marketing use with one of the individual components. The adverse reactions reported for ezetimibe/simvastatin are consistent with those previously reported with ezetimibe and/or simvastatin.

Investigations: liver function test abnormal

Blood and lymphatic system disorders: thrombocytopenia; anemia

Nervous system disorders: peripheral neuropathy; myasthenia gravis

Eye disorders: ocular myasthenia

Respiratory, thoracic and mediastinal disorders: cough; interstitial lung disease

Gastrointestinal disorders: constipation; pancreatitis; gastritis

Skin and subcutaneous tissue disorders: alopecia; hypersensitivity reactions, including rash, lichenoid drug eruptions, urticaria, anaphylaxis, angio-edema; severe cutaneous adverse reactions (SCARs), including Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilic and systemic symptoms (DRESS) and erythema multiforme

Musculoskeletal and connective tissue disorders: muscle cramps; myopathy/rhabdomyolysis

There have been very rare reports of immune-mediated necrotizing myopathy (IMNM), an autoimmune myopathy, associated with statin use. IMNM is clinically characterized by: persistent proximal muscle weakness and elevated serum creatine kinase, which persist despite discontinuation of statin treatment; muscle biopsy showing necrotizing myopathy without significant inflammation; improvement with immunosuppressive agents.

Metabolism and nutrition disorders: decreased appetite

Vascular disorders: hot flush; hypertension

General disorders and administration site conditions: pain

Hepato-biliary disorders: hepatitis/jaundice; fatal and non-fatal hepatic failure; cholelithiasis; cholecystitis; drug-induced liver injury

Reproductive system and breast disorders: erectile dysfunction

Psychiatric disorders: depression

An apparent hypersensitivity syndrome has been reported rarely which has included some of the following features: angioedema, lupus-like syndrome, polymyalgia rheumatica, dermatomyositis, vasculitis, thrombocytopenia, eosinophilia, ESR increased, arthritis and arthralgia, urticaria, photosensitivity, fever, flushing, dyspnea and malaise.

There have been rare post marketing reports of cognitive impairment (e.g., memory loss, forgetfulness, amnesia, memory impairment, confusion) associated with statin use. These cognitive issues have been reported for all statins. The reports are generally nonserious, and reversible upon statin discontinuation, with variable time to symptom onset (1 day to years) and symptom resolution (median of 3 weeks).

Laboratory Values

In controlled clinical coadministration trials, the incidence of clinically important elevations in serum transaminases (ALT and/or AST ≥ 3 X ULN, consecutive) was 1.7% for patients treated with ezetimibe/simvastatin. These elevations were generally asymptomatic, not associated with cholestasis, and returned to baseline after discontinuation of therapy or with continued treatment.

Clinically important elevations of CK (≥ 10 X ULN) were seen in 0.2% of the patients treated with ezetimibe/simvastatin.

Increases in HbA1c and fasting serum glucose levels have been reported with statins, including simvastatin. The risk of hyperglycemia, however, is outweighed by the reduction in vascular risk with statins.

OVERDOSAGE

XATHOR

No specific treatment of overdosage with XATHOR can be recommended. In the event of an overdose, symptomatic and supportive measures should be employed.

Ezetimibe

A few cases of overdosage have been reported; most have not been associated with adverse experiences. Reported adverse experiences have not been serious.

Simvastatin

A few cases of overdosage have been reported; the maximum dose taken was 3.6 g. All patients recovered without sequelae.

STORAGE

Store below 30°C. Keep in a dry place.

AVAILABILITY

In box of 30 tablets (3 x 10's) in blister of Alu/Alu or PVC Aclar/Alu.

SHELF LIFE

Product should not be used beyond the expiry date imprinted on the product packaging.

PRODUCT REGISTRATION HOLDER/MANUFACTURER:
PHARMANIAGA MANUFACTURING BERHAD (198001006232)
No.11A, Jalan P/I, Kawasan Perusahaan Bangi,
43650 Bandar Baru Bangi, Selangor Darul Ehsan, Malaysia

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PRP XXXXXXXXXX